

PROJECT REPORT

Sentiment Analysis System

Internship Project Report

Submitted By

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Acknowledgement

I sincerely thank Transen Dynamics for providing me with the opportunity to work on this Sentiment Analysis System internship project. This project helped me understand the practical implementation of Machine Learning and Natural Language Processing techniques.

I also express my gratitude to my college faculty and mentors for their continuous guidance and support throughout the project.

Abstract

The Sentiment Analysis System is a Machine Learning application developed to classify movie reviews into positive or negative sentiments. The project uses the IMDb Movie Reviews Dataset containing 50,000 labeled reviews. "This implementation performs binary

sentiment classification (positive and negative) because the IMDb 50K dataset does not contain neutral labels.

1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses a positive, negative, or neutral opinion.

Organizations use sentiment analysis to analyze customer feedback, product reviews, social media posts, and survey responses.

This internship project demonstrates the implementation of a sentiment analysis system using Machine Learning techniques.

2. Project Overview

The objective of this project is to develop a system capable of automatically identifying the sentiment of movie reviews.

The system performs:

- Dataset collection
- Data preprocessing
- Feature extraction
- Machine Learning model training
- Performance evaluation
- User prediction

3. Objectives

The objectives of this project are:

- Develop a sentiment analysis system.
- Classify reviews based on sentiment.
- Clean and preprocess textual data.
- Train a Machine Learning model.
- Evaluate the model using performance metrics.
- Provide users with an interface to predict sentiment.

4. Dataset Collection

The dataset used for this project is:

IMDb 50K Movie Reviews Dataset

Source:

<https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>

Dataset Details

- Total Reviews: 50,000
- Positive Reviews: 25,000
- Negative Reviews: 25,000
- File Format: CSV

Columns:

Column	Description
review	Movie review text
sentiment	Positive or Negative

5. Data Preprocessing

The preprocessing stage included:

- Converting text to lowercase
- Removing punctuation
- Removing stop words
- Cleaning unnecessary characters

These preprocessing steps improve the quality of text before feature extraction.

6. Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert textual reviews into numerical vectors.

Advantages:

- Reduces the importance of common words.
- Highlights meaningful words.
- Improves model performance.

7. Machine Learning Model

The Logistic Regression algorithm was selected because it:

- Performs well for text classification.
- Is computationally efficient.
- Produces high accuracy.
- Is simple to interpret.

Pipeline Used

Text Review

↓

Preprocessing

↓

TF-IDF Vectorizer

↓

Logistic Regression

↓

Sentiment Prediction

8. System Implementation

The implementation consists of:

- Loading the IMDb dataset
- Preprocessing text
- Splitting data into training and testing sets
- Training Logistic Regression model
- Evaluating the model
- Saving trained model
- Accepting user reviews
- Predicting sentiment

Programming Language:

Python

Libraries Used:

- Pandas
- NumPy
- Scikit-learn
- Joblib

9. Model Evaluation

Performance Metrics Used

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Result

Accuracy Achieved

89.52%

Classification Report

Precision : 0.90

Recall : 0.90

F1 Score : 0.90

Confusion Matrix

```
[[4351 610]
```

```
[438 4601]]
```

Accuracy: 0.8952

Classification Report:

	precision	recall	f1-score	support
negative	0.91	0.88	0.89	4961
positive	0.88	0.91	0.90	5039
accuracy			0.90	10000
macro avg	0.90	0.90	0.90	10000
weighted avg	0.90	0.90	0.90	10000

Confusion Matrix:

```
[[4351  610]
 [ 438 4601]]
```

Model Saved Successfully!

The model demonstrates high prediction performance on unseen movie reviews.

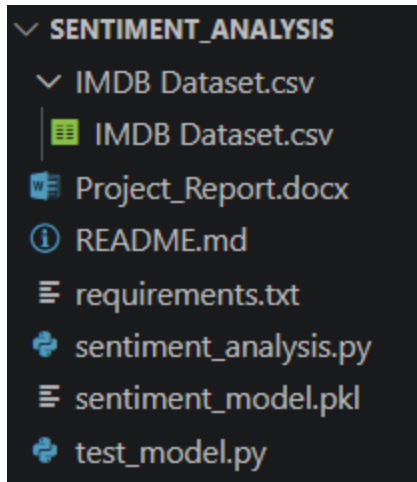
10. Sample Input and Output

```
Enter Review (type 'exit' to quit): This movie was absolutely amazing. I loved every moment.  
Predicted Sentiment: positive  
  
Enter Review (type 'exit' to quit): This was the worst movie I have ever seen.  
Predicted Sentiment: negative  
  
Enter Review (type 'exit' to quit): Waste of time. Terrible acting and boring story.  
Predicted Sentiment: negative  
  
Enter Review (type 'exit' to quit): exit  
Thank You!  
PS C:\Users\mikke\OneDrive\Desktop\Sentiment_Analysis>
```

11. Project Deliverables

Completed deliverables include:

- Python Source Code
- IMDb Dataset
- Trained Machine Learning Model
- README Documentation
- Requirements File
- Project Report



12. Advantages

- High prediction accuracy
- Fast classification
- Easy to use
- Scalable
- Efficient text preprocessing

13. Limitations

- Supports only positive and negative sentiment because the IMDb dataset contains only two sentiment classes.
- Performance depends on dataset quality.
- Does not detect sarcasm or mixed emotions.

14. Future Scope

Future improvements include:

- Support neutral sentiment classification.

- Develop a web application using Flask.
- Deploy the model on cloud platforms.
- Improve prediction using Deep Learning models such as LSTM or BERT.
- Add multilingual sentiment analysis.

15. Conclusion

The Sentiment Analysis System was successfully developed using Machine Learning techniques.

The system preprocesses textual data, converts text into numerical features using TF-IDF, and predicts sentiment using Logistic Regression.

The trained model achieved an accuracy of **89.52%**, demonstrating reliable performance on the IMDb dataset. The project fulfills the internship objectives of data collection, preprocessing, model development, evaluation, implementation, and documentation.